



Reading: Exemplar: Maintain efficient process utilization on Windows

Introduction

In this lab, you used the new commands you learned to do some process maintenance on a Windows virtual machine. As an IT Support Specialist, it's super important that you maintain efficient process utilization on your machines.

This exemplar is a walk-through of the previous Qwiklab activity, including detailed instructions and solutions. You may use this exemplar if you were unable to complete the lab and/or you need extra guidance in competing lab tasks. You may also refer to this exemplar to prepare for the graded quiz in this module.

Scenario

There are three primary components to this lab:

- Collect process information using the Task Viewer.
- Terminate a specific process using Windows PowerShell.
- Terminate multiple processes using Windows PowerShell.

Start the lab

You'll need to start the lab before you can access the materials. To do this, click the green **Start Lab** button at the top of the screen.

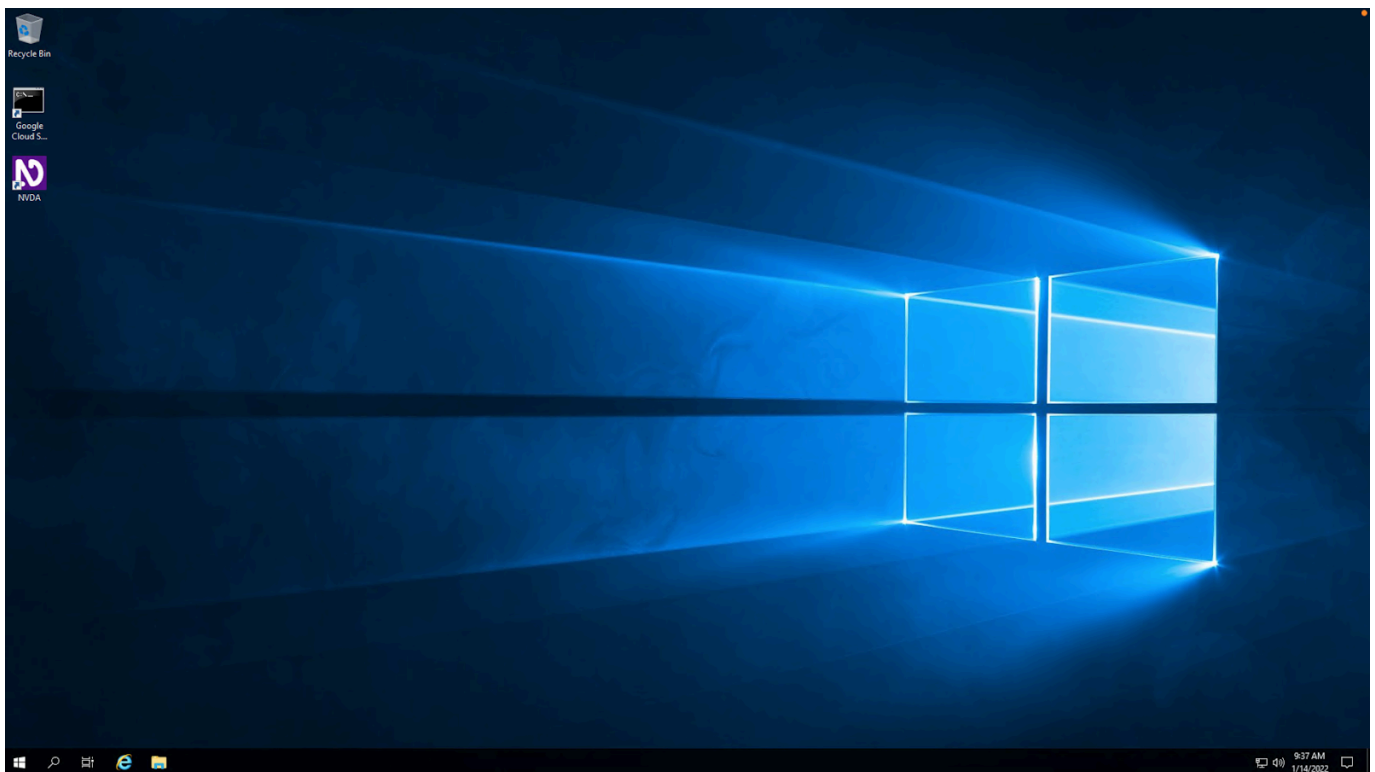
A green rectangular button with the text "Start Lab" in white.

After you click the **Start Lab** button, you will see a panel appear below where the start lab button was that has an **Open Windows VM** button.

Caution: When you are in the console, do not deviate from the lab instructions. Doing so may cause your account to be blocked.
[Learn more.](#)

Open Windows VM

Click the **Open Windows VM** button and a new tab will open with a visual interface for Windows OS, where you will be performing further steps in the lab. You should have a visual interface for Windows that looks like this:



Terminating a specific process

On Windows, you can view running processes in the Task Viewer, or use Windows PowerShell (this is what you'll be using for this lab). For these operations, you'll need to be running a Windows PowerShell terminal in **Administrative** mode. So, search the **Start Menu** for **Windows PowerShell**, right-click it, and select **Run as Administrator**.

From Windows PowerShell, you can use **Get-Process** to search for a process by name. The **totally_not_malicious** process is running on this machine, too. Search for it, using this command:

```
Get-Process -Name "totally_not_malicious"
```

Each row represents a process, and one of the columns shows the process ID.

OUTPUT:

Handles	NPM(K)	PM(K)	WS(K)	CPU(s)	Id	SI	ProcessName
-----	-----	-----	-----	-----	--	--	-----
209	14	2724	10996	271.06	7164	1	totally_not_malicious

To end a process, you can use **taskkill** and specify the Process ID, or PID, of the process:

Note: Make sure you replace/substitute the **[PROCESS ID]** with the ID of the process you got from the previous command. Here is an example of how to type the command with the process ID: **taskkill /F /PID 2156**

```
taskkill /F /PID [PROCESS ID]
```

You should see this message after running **taskkill** with the PID for your process, which will likely be different than the ID specified here:

OUTPUT:

SUCCESS: The process with PID 7164 has been terminated.

To verify that the process is no longer running, you can search for it again:

```
Get-Process -Name "totally_not_malicious"
```

This should throw an error because no process by that name exists anymore, indicating that you've successfully ended it:

OUTPUT:

```
Get-Process : Cannot find a process with the name "totally_not_malicious". Verify the process
name and call the cmdlet
again.
At line:1 char:1
+ Get-Process -Name "totally_not_malicious"
+ ~~~~~
+ CategoryInfo          : ObjectNotFound: (totally_not_malicious:String) [Get-Process],
ProcessCommandException
+ FullyQualifiedErrorId :
NoProcessFoundForGivenName,Microsoft.PowerShell.Commands.GetProcessCommand
```

Terminate multiple processes

There are processes containing the word "razzle" also running on this VM. **Get-Process** doesn't handle processes with partially-matching names, like **grep** does, and running **Get-Process -Name "razzle"** would result in no matches. However, you can use "wildcards" (asterisks) to look for processes that contain "razzle" in their name:

```
Get-Process -Name "*razzle*"
```

This will show two processes that contain "razzle" in their name.

OUTPUT:

Handles	NPM(K)	PM(K)	WS(K)	CPU(s)	Id	SI	ProcessName
-----	-----	-----	-----	-----	--	--	-----
209	14	2784	11028	572.89	2120	1	my_cat_razzle
209	14	2824	11036	572.83	7052	1	razzle_dazzle

You can use **taskkill**, like before, once for each of the "razzle" processes:

Note: Make sure you replace/substitute the **[PROCESS ID]** with the ID of the process you got from the previous command.

```
taskkill /F /PID [PROCESS ID]
```

OUTPUT:

```
PS C:\users\qwiklabs> taskkill /F /PID 2120
SUCCESS: The process with PID 2120 has been terminated.
PS C:\users\qwiklabs> taskkill /F /PID 7052
SUCCESS: The process with PID 7052 has been terminated.
```

You can use **Get-Process** again to verify that the processes have been ended:

```
Get-Process -Name "*razzle*"
```

You shouldn't see any processes in the output. When you ran this before to verify that the malicious process had been terminated, it printed an error message because the specifically-named process was not present. When you use a wildcard (*) in the search, you aren't looking for an exact match. So, rather than an error message, the command outputs nothing at all (because there are no matches):

```
PS C:\users\qwiklabs> Get-Process -Name "*razzle*"
PS C:\users\qwiklabs>
```

Conclusion

Congratulations! You've successfully used the Windows PowerShell commands **Get-Process** to find Windows processes, and **taskkill** to end them. As an IT Support Specialist, it's important for you to monitor system processes and maintain them using the Task Viewer and Windows PowerShell.
